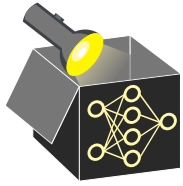
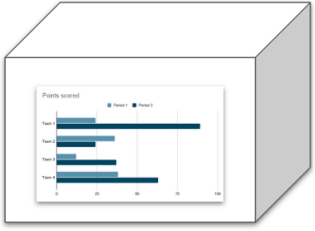
Interpretable Machine Learning



Interpretable Models
Inherently Interpretable Models - Motivation

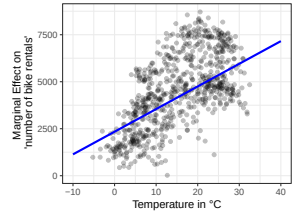


Learning goals

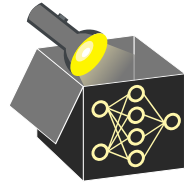
- Why should we use interpretable models?
- Advantages and disadvantages of interpretable models

MOTIVATION

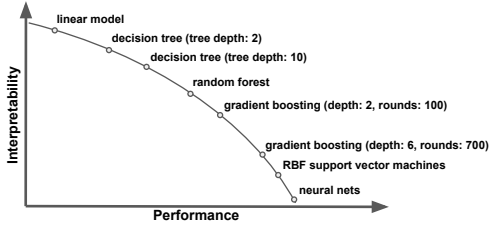
- Achieving interpretability by using interpretable models is the most straightforward approach
- Classes of models deemed interpretable:
  - (Generalized) linear models (LM, GLM)
  - Generalized additive models (GAM)
  - Decision trees
  - Rule-based learning
  - Model-based / component-wise boosting
  - ...



$\rightsquigarrow$ LM provides straightforward interpretation

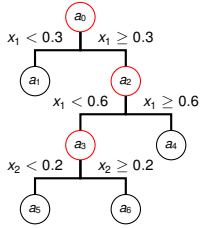


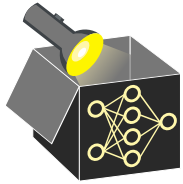
- Often there is a trade-off between interpretability and model performance



ADVANTAGES

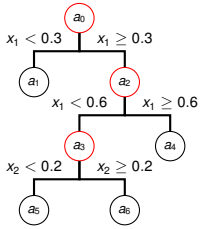
- Interpretable models are transparent by design, making many model-agnostic explanation methods unnecessary
  $\rightsquigarrow$ Eliminates an extra source of estimation error

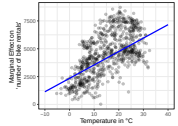


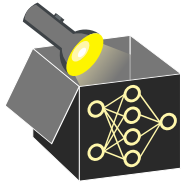


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- Interpretable models are transparent by design, making many model-agnostic explanation methods unnecessary
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- They often have few hyperparameters and are structurally simple (e.g., linear, additive, sparse, monotonic)
  $\rightsquigarrow$ Easy to train, fast to tune, straightforward to explain

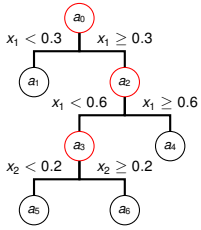


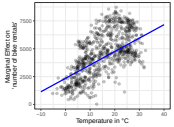


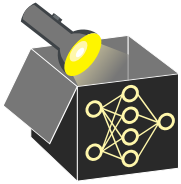


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  $\rightsquigarrow$ Easy to train, fast to tune, straightforward to explain
- Many people are familiar with simple interpretable models
  $\rightsquigarrow$ Increases trust, facilitates communication of results

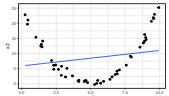


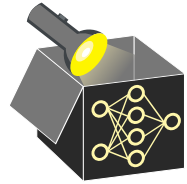




DISADVANTAGES & LIMITATIONS

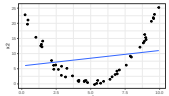
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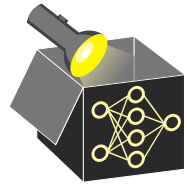




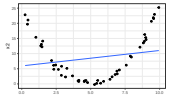
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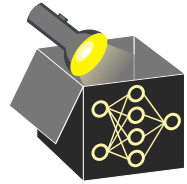
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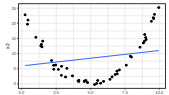
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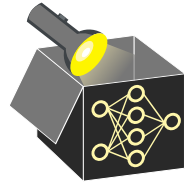




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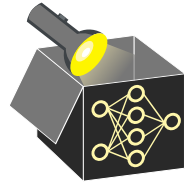
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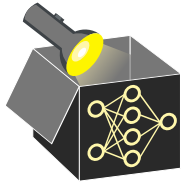
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- Often do not automatically model complex relationships due to limited flexibility
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- Inherently interpretable models do not address all explanation needs
  $\rightsquigarrow$ Complementary model-agnostic methods (e.g., counterfactuals) remain valuable for specific tasks

FURTHER COMMENTS



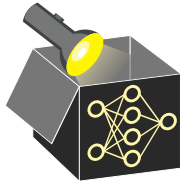
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  - Built-in interpretation
    ⇝ fewer risks from misleading post-hoc explanations
  - Good performance possible with effort on preprocessing and/or feature engineering
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  - But interpretability depends on meaning of created features
    $\rightsquigarrow$ E.g., PCA keeps models linear, but yields hard-to-interpret components
- Limitation: Less suited for complex data requiring end-to-end learning
  - Applies to image, text, or sensor data where features must be learned from raw input
  - Manual extraction of interpretable features is difficult
    $\Rightarrow$ Information loss and lower performance

RECOMMENDATION



- Begin with the simplest model appropriate for the task
- Increase complexity only if necessary to meet performance requirements
  $\rightsquigarrow$ Typically reduces interpretability and requires model-agnostic explanations
- Choose the simplest model with sufficient accuracy
  $\rightsquigarrow$ Occam's razor

Bike Data, 4-fold CV

Model	RMSE	R^2
LM	800.15	0.83
Tree	981.83	0.74
Random Forest	653.25	0.88
Boosting (tuned)	638.42	0.89